Collapse all

Welcome

Complete all items

[Step 1: Start Here]
Mark completed

[Step 2: Enroll in Discussion Forum]
Mark completed

[Step 3: Check Your Canvas Communication Settings]
Mark completed

[Step 4: Proctoring Information]
Mark completed

[Step 5: ID Verification Onboarding Quiz]
0 pts Submit

[Step 6: Prerequisites Quiz]
0 pts Submit

Course Information

Prerequisites: Welcome

Welcome Video

Meet Your Instructor

Help and Support

Module One

Prerequisites: Welcome

Complete all items

Lesson One: What is Network Science?

Lesson One Overview
View

L1: What Is Network Science?
View

L1: Complex Systems
View

L1: Trivial Networks Versus Complex Networks
View

L1: Example: The Brain of the C.elegans Worm
View

L1: The Main Premise
View

L1: Examples of Systems Studied by Network Scientists
View

L1: Network Centrality
View

L1: Communities (Modules) in Networks
View

L1: Dynamics of Networks
View

L1: Influence and Cascade Phenomena
View

L1: Machine Learning and Network Science
View

L1: The History of Network Science
View

L1: The Birth of Network Science
View

L1: Lesson Summary
View

L1: TED Lecture: Albert-László Barabási
View

L1: Knowledge Check
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Lesson Two: Relevant Concepts From Graph Theory

Lesson Two Overview
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L2: An Introduction
View

L2: Undirected Graphs

View

L2: Adjacency Matrix
View

L2: Adjacency List
View

L2: Walks, Paths and Cycles
View

L2: Trees and Other Regular Networks
View

L2: Directed Graphs
View

L2: Weighted Directed Graphs
View

L2: (Weakly) Connected Components
View

L2: Strongly Connected Components
View

L2: Directed Acyclic Graphs (DAGs)
View

L2: Dijkstra’s Shortest Path Algorithm
View

L2: Random Walks

L2: Min-Cut Problem
View

L2: Max-flow Problem

L2: Max-flow=Min-cut

L2: Bipartite Graphs
View

L2: A Recommendation System as a Bipartite Graph
View

L2: Co-citation and Bibliographic Coupling
View

L2: Lesson Summary
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L2: Knowledge Check
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Module Two

Prerequisites: Welcome Complete all items

Lesson Three: Degree Distribution and The "Friendship Paradox"

Lesson Three Overview
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L3: Degree Distribution
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L3: Degree Distribution Moments
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L3: Two Special Degree Distributions
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L3: Example: Degree Distribution of a Sex-Contact Network
View

L3: Friendship Paradox
View

L3: Two Extreme Cases of The Friendship Paradox
View

L3: Application of Friendship Paradox in Immunization Strategies
View

L3: The G(n,p) model (ER Graphs)
View

L3: Degree Distribution of G(n,p) Model
View

L3: Connected Components in G(n,p)
View

L3: Size of LCC in G(n,p) as Function of p
View

L3: When Does G(n,p) have a Single Connected Component?
View

L3: Degree Correlations
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L3: How to Measure Degree Correlations
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L3: Assortative, Neutral and Disassortative Networks
View

L3: Lesson Summary
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L3: Knowledge Check
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Lesson Four: Random vs. Real Graphs and Power-Law Networks

Lesson Four Overview
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L4: Degree Distribution of Real Networks
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L4: Power-law Degree Distribution
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L4: The Role of the Exponent of a Power-law Distribution
View

L4: How to Check if a Network Has Power-law Degree Distribution
View

L4: How to Plot Power-law Degree Distribution
View

L4: Scale-free Nature of Power-law Networks
View

L4: The Maximum Degree in a Power-law Network
View

L4: Random Failures and Targeted Attacks in Power-law Networks
View

L4: Degree-preserving Randomization
View

L4: The Average Degree of the Nearest Neighbor at a Power-law Network
View

L4: Configuration Model
View

L4: Preferential Attachment Model

View

L4: Link Selection Model

View

L4: What Does The Power-law Property Mean in Practice?

View

L4: Case Studies: Superspreaders

L4: Lesson Summary

L4: Knowledge Check

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Lesson Five: Network Paths, Clustering and The “Small World” Property

Lesson Five Overview

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L5: Clustering Coefficient

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L5: Average Clustering and Transitivity Coefficient

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L5: Clustering in Weighted Networks

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L5: Clustering in G(n,p) networks

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L5: Clustering in Regular Networks

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L5: Diameter, Characteristic Path Length, and Network Efficiency

View

L5: Diameter and CPL of G(n,p) Networks

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L5: Diameter and Efficiency of G(n,p) Versus Regular Networks

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L5: What Does “Small-world” Network Mean?

View

L5: Clustering in PA Model

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L5: Length of Shortest Paths in PA Model

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L5: Path Lengths in Power-law Networks
View

L5: Directed Subgraph Connectivity Patterns
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L5: Statistical Test For The Frequency of a Network Motif
View

L5: Frequent Motifs and Their Function

L5: Knowledge Check
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Module Three

Prerequisites: Welcome Complete all items

Lesson Six: Centrality and Network-core Metrics and Algorithms

Lesson Six Overview
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L6: Degree, Eigenvector and The Katz Centrality
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L6: PageRank Centrality
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L6: Closeness Centrality and Harmonic Centrality
View

L6: Betweenness Centrality Variants
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L6: Path Centrality For Directed Acyclic Graphs
View

L6: The Notion of "Node Importance"
View

L6: k-core Decomposition
View

L6: Core-Periphery Structure
View

L6: Rich-Club Set of Nodes
View

L6: Core Set of Nodes in DAGs
View

L6: Applications
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L6: Lesson Summary

L6: Knowledge Check
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Lesson Seven: Modularity and Community Detection

Lesson Seven Overview
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L7: The Graph Partitioning Problem
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L7: Network Community Detection

L7: Community Detection Based on Edge Centrality Metrics
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L7: Divisive Hierarchical Community Detection
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L7: Agglomerative Hierarchical Clustering Approaches - Node Similarity Metric
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L7: Hierarchical Clustering Approaches – Group Similarity
View

L7: Modularity Metric
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L7: Modularity Metric- Derivation
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L7: Modularity Metric – Selecting The Community Assignment
View

L7: Modularity After Merging Two Communities
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L7: Greedy Modularity Maximization
View

L7: Louvain Algorithm
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L7: Modularity Resolution

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L7: The Modularity Search Landscape
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L7: Communities Within Communities
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L7: Clustering Coefficient Versus Degree in Practice
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L7: Hierarchical Modularity Through Recursive Community Detection
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L7: Lesson Summary
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Lesson Eight: Advanced Topics in Community Detection

Lesson Eight Overview
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L8: Overlapping Communities and CFinder Algorithm
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L8: Critical Density Threshold in CFinder
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L8: Link Clustering Algorithm
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L8: Application of Link Clustering Algorithm
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L8: GN Benchmark
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L8: Community Size Distribution
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L8: LFR Benchmark
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L8: Normalized Mutual Information (NMI) Metric
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L8: Accuracy Comparison of Community Detection Algorithms
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L8: Run Time Comparisons
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L8: Dynamic Communities
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L8: Dynamic Communities – Approach #1
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L8: Dynamic Communities – Approach #2
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L8: Dynamic Communities – Approach #3
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L8: Participation Coefficient
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L8: Within-module Degree
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L8: Case Study: Metabolic Networks
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L8: δ-MAPS

L8: δ-MAPS – Domain Homogeneity Constraint
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L8: δ-MAPS: Algorithm Overview
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L8: Application in Climate Science
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L8: Lesson Summary
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L8: Knowledge Check
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Module Four

Prerequisites: Welcome Complete all items

Lesson Nine: Spreading Phenomena on Networks and Epidemics

Lesson Nine Overview
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L9: Spreading Phenomena on Networks & Epidemics
View

L9: SI Model
View

L9: SIS Model
View

L9: SIR Model
View

L9: Comparison of SI, SIS, SIR Models Under Homogeneous Mixing
View

L9: Number of Partners in Sexual Networks
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L9: Number of “Close Proximity” Contacts
View

L9: Global Travel Network
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L9: Reproductive Number R0
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L9: The Fallacy of The Basic Reproductive Number
View

L9: Degree Block Approximation
View

L9: SIS Model – With An Arbitrary Degree Distribution
View

L9: SIS Model – No Epidemic Threshold For Scale-Free Nets
View

L9: Summary of SI, SIS, SIR Models with Arbitrary Degree Distribution
View

L9: Computational Modeling of Epidemics
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L9: Effective Distance
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L9: Lesson Summary
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Lesson Ten: Social Influence and Other Network Contagion Phenomena

Lesson Ten Overview

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L10: Not Just Viruses Spread On Networks

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L10: Diffusion, Cascades and Adoption Models

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L10: Some Empirical Findings About Cascades

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L10: Linear Threshold Model

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L10: Independent Contagion Model

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L10: Deffuant Model For Opinion or Consensus Formation

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L10: Game Theoretic Diffusion Models

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L10: Seeding for Maximum Network Cascade

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L10: Submodularity of Objective Function

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L10: Monotone and Submodular Function

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L10: Cascades in Networks with Communities

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L10: Cascades in Networks with Communities-2

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L10: How do Dense Clusters Affect Cascades?

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L10: Network Diffusion in the Brain

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L10: Asynchronous Linear Threshold (ALT) Model

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L10: Experimental Validation of ALT Model

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L10: Lesson Summary

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L10: Knowledge Check
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Lesson Eleven: Other Network Dynamic Processes

Lesson Eleven Overview
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L11: Inverse Percolation On Grids and G(n,p) Networks
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L11: Random Removals (Failures) vs Targeted Removals (Attacks)
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L11: Molloy-Reed Criterion
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L11: Robustness of Networks to Random Failures
View

L11: Robustness of Networks to Attacks
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L11: Small-world Networks and Decentralized Search
View

L11: Decentralized Search Problem
View

L11: The Optimal Search Exponent in Two Dimensions

L11: Why The Optimal Value of q is Equal to Two in a Two-Dimensional World?

L11: Decentralized Network Search in Practice

L11: Synchronization of Coupled Network Oscillators
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L11: Coupled Phase Oscillators – Kuramoto Model
View

L11: Kuramoto Model on a Complete Network
View

L11: Kuramoto Model on Complex Networks
View

L11: Adaptive (or Coevolutionary) Networks
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L11: Consensus Formation in Adaptive Networks

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L11: Coevolutionary Effects in Twitter Retweets
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L11: Lesson Summary
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L11: Knowledge Check
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Module Five

Prerequisites: Welcome Complete all items

Lesson Twelve: Network Modeling

Lesson Twelve Overview
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L12: Why Network Modeling?
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L12: Preferential Attachment Model
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L12: Mathematical Analysis of PA Model
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L12: Degree Dynamics in PA Model
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L12: Nonlinear Preferential Attachment
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L12: Link-Copy Model
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L12: Generating Networks with Community Structure
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L12: Generating Networks with Degree Correlations
View

L12: Optimization-based Network Formation Model
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L12: Optimization-based Network Formation Model (cont')
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L12: Hierarchical Graph Model
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L12: Maximum Likelihood Estimation of HRG Probabilities
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L12: How to Find the Optimal Dendrogram?
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L12: Hierarchical Graph Model Applications
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L12: Lesson Summary
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Lesson Thirteen: Statistical Analysis of Network Data

Lesson Thirteen Overview
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L13: Introduction to Network Sampling
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L13: Network Sampling Strategies
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L13: Inclusion Probabilities with Each Sampling Strategy
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L13: Horvitz-Thompson Estimation of "Node Totals"
View

L13: Estimating the Number of Edges and the Average Degree
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L13: Estimating the Number of Nodes with Traceroute-like Methods
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L13: Topology Inference Problems
View

L13: Link Prediction
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L13: Association Networks
View

L13: Topology Inference Using Network Tomography
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L13: Other Network Estimation and Tomography Problems

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L13: Lesson Summary
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L13: Knowledge Check
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Lesson Fourteen: Machine Learning in Network Science

Lesson Fourteen Overview
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L14: Embedding Nodes in a Low-dimensional Space
View

L14: Shallow Encodings and Similarity Metrics
View

L14: Random-walk Encodings and Node2vec
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L14: Applications of Node Embeddings
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L14: Deep Embeddings and Graph Neural Networks
View

L14: GNN Training and Decoders
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L14: Application: Polypharmacy
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L14: Advanced Topics: Deep Generative Models for Graphs
View

L14: Advanced Topics: Interdependent Networks
View

L14: Advanced Topics: Temporal Networks
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L14: Lesson Summary
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L14: Knowledge Check
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A Final Note